A Semi-Automated Approach to Analyzing Natural Conversations in Autistic Youth: A Pilot Study

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Introduction

Natural conversations can show how autistic youth communicate through the words they use and the length of their spoken phrases and sentences. Manual transcription and scoring take considerable time, while automated transcription can introduce errors [1].

This pilot explored whether a semi-automated software system with brief human review could reliably measure language characteristics from natural conversations with autistic youth while reducing the amount of manual analysis required.

Methodology

Participants and recordings

10 autistic participants, ages 8–17
37 conversations, about 8 minutes each
2–4 recordings per participant

What we measured

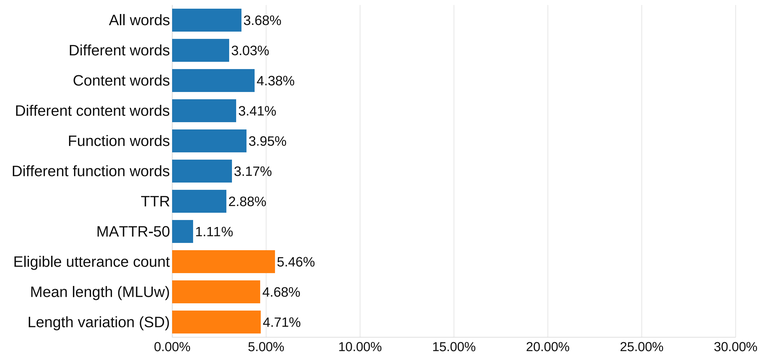
The system calculated 11 measures of vocabulary use, diversity, and utterance length. These included type–token ratio (TTR), moving-average TTR over 50 words (MATTR-50), and mean words per utterance (MLUw).

How we checked accuracy

We compared uncorrected pipeline measures with TAALED and CLAN measures from corrected text using the same utterance rules [2,3], and with about 100 human-labeled utterances per participant (14 recordings). Two additional reviewers each labeled two recordings to assess boundary agreement.

Comparison against TAALED and CLAN

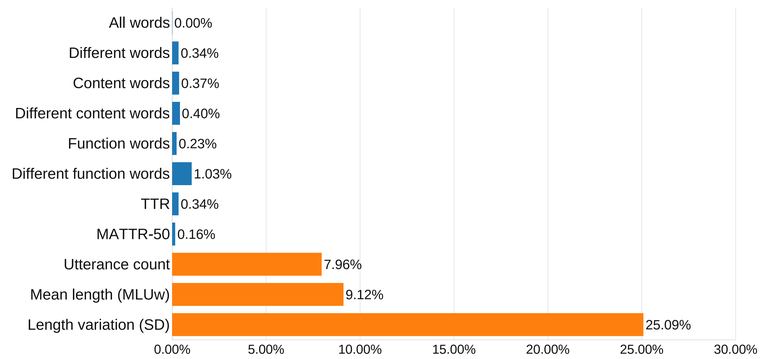
37 recordings (Mean Absolute Percentage Error): Our system vs. corrected TAALED/CLAN results.



MAPE. Blue: vocabulary measures. Orange: utterance measures.

Comparison with 100-utterance samples

14 recordings (Mean Absolute Percentage Error): About 100 manually labeled utterances per participant.



Lower error means closer agreement.

Boundary Agreement

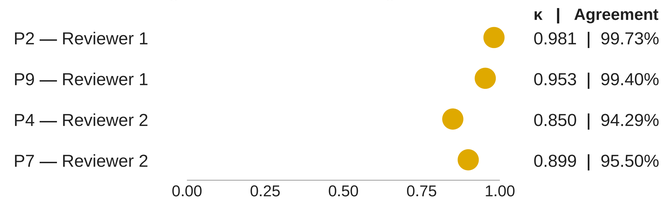
Reviewer 1 and Reviewer 2 each labeled two recordings. We compared utterance boundaries at aligned word gaps, retaining fillers [4].

0.935
Cohen's κ

98.70%
Boundary agreement

Pooled across 2,777 evaluated word gaps

Cohen's κ adjusts agreement for chance.
0 = chance-level agreement; 1 = complete agreement.



Dots closer to 1 mean stronger agreement beyond chance.

Discussion

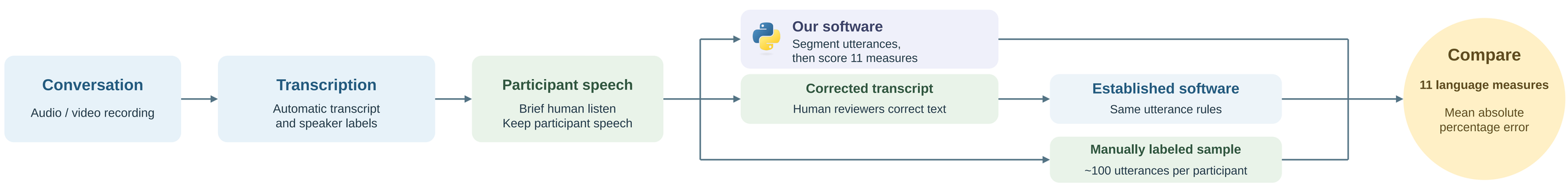
Vocabulary MAPE was 1.11–4.38% against TAALED and 0.00–1.03% against the labeled samples.

Both transcript versions used the same utterance rules. MLUw error was 4.68% against corrected CLAN transcripts and 9.12% against the labeled samples.

Fillers remain in utterance counts and boundary comparisons. Word metrics exclude fillers. CLAN and our MLU calculator agreed on identical inputs.

This descriptive pilot supports further research with independent samples and recording review. It does not establish diagnostic or clinical validity.

System Diagram



References

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Error is MAPE, calculated within each participant, then averaged equally across participants.